

DIYA SHETTY

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EDUCATION

North Carolina State University – Raleigh, NC

May 2026

Master of Computer Science (MCS) | GPA: 4.0/4.0

Coursework: Data Structures & Algorithms, Software Engineering, Database Systems, Foundations of Data Science, Computer Networks

BMS College of Engineering – Bengaluru, India

May 2024

Bachelor of Engineering (B.E.) – Information Science and Engineering

TECHNICAL SKILLS

- **Languages:** Java, Python, JavaScript, TypeScript, SQL, C++, HTML, CSS, R
- **Frameworks & Libraries:** Spring Boot, Spring Security, FastAPI, Next.js, React.js, Node.js, LangChain, TensorFlow, Pandas, NumPy
- **Cloud & Databases:** AWS (Glue, S3), Azure (Digital Twins), PostgreSQL, MySQL, MongoDB, ChromaDB, Docker, Linux, Git, CI/CD
- **Concepts & Tools:** REST APIs, JWT Auth, OOP, System Design, Agile/Scrum, DSA, RAG, Agentic AI, Wireshark

WORK EXPERIENCE

Data Analyst Intern · Genpact – New York, USA

Jun 2025 – Aug 2025

- Designed and deployed an **AI-powered** Market Scope Analysis platform using Agentic AI, RAG, and LLM-based orchestration; built autonomous agents for web scraping, document retrieval, and vector search, cutting manual research time by **~60%**.
- Built an AWS Glue ETL pipeline with dimensional modeling (star schema) to transform raw data into fact/dimension tables, enabling automated daily analysis of **10K+ records** across multiple business domains.

Summer Intern · EY – Chennai, India

Mar 2024 – Apr 2024

- Designed an SAP-ERP business blueprint for a simulated retail business across 6 core modules (procurement, planning, manufacturing, sales & distribution); approved by manager for demonstrating end-to-end process understanding.
- Built a multi-module ERP demo application in HTML, CSS, JavaScript, and Node.js, translating the blueprint into functional end-to-end workflows.

Technical Operations Intern · Unfold Consulting – Bangalore, India

Aug 2023 – Nov 2023

- Executed 250+ UAT test cases across dev and production environments, enabling early go-live on select features
- Identified a critical defect in the production environment, along with 10+ other database and data-upload issues, preventing potential post-launch failures; retested fixes alongside developers to validate resolution before deployment.

PROJECTS

Document Q&A – Full-Stack RAG Application

- Architected a full-stack RAG application with a FastAPI (Python) backend and Next.js/TypeScript frontend, handling multi-document PDF uploads of 15-30 pages with fast, real-time query responses.
- Designed a normalized PostgreSQL schema (users, documents, conversations, messages) via SQLAlchemy ORM; implemented NextAuth session-token middleware forwarding credentials to FastAPI for secure, end-to-end request authentication
- Delivered FastAPI StreamingResponse endpoints (/upload, /ask) and a real-time React/TypeScript chat UI featuring PDF upload with progress tracking, per-document chat history, and protected routes with NextAuth session guards.

IoT – Azure Cloud Digital Twin

 Best Innovation, Ingenius 2024

- Architected an IoT monitoring solution using Azure Digital Twins (ADT), Azure Data Factory, and REST APIs to stream real-time device telemetry, reducing manual monitoring effort by 30%.
- Designed and exposed 3 RESTful API endpoints connecting ADT models to a cross-platform mobile app, delivering live sensor data, health predictions, and anomaly alerts to end users.
- Built an interactive 3D visualization with dynamic health overlays and time-series graphs; implemented real-time alert logic to flag abnormalities and trigger notifications.

Cryptography & Secure Networking

- Implemented a secure file transfer protocol in Python with AES-256-GCM encryption, PBKDF2 key derivation, and Diffie-Hellman key exchange, tested across multiple file sizes and types; used PyCryptodome and socket programming to build custom PDUs.
- Engineered HMAC-SHA256 integrity verification to ensure tamper detection and authenticity validation across all transmitted data.
- Developed a man-in-the-middle proxy that consistently exposed Diffie-Hellman key exchange vulnerabilities, validating real-world threat models in a controlled test environment

Natural Language Processing – IEEE Published

ieeexplore.ieee.org/document/10544307

- Fine-tuned Bio+Clinical BERT on 3+ NLP classification tasks for healthcare applications, achieving 99% accuracy across named entity recognition and clinical text categorization; published in IEEE (DOI: 10.1109/ICCIIT60459.2023.10544307).
- Applied LIME and SHAP explainability frameworks to evaluate 10 model predictions, surfacing per-feature contributions and enabling transparent model debugging for clinical use cases.
- Diagnosed class imbalance across healthcare datasets and derived resampling and threshold-tuning strategies to improve recall on minority classes and overall pipeline fairness.